Series of real numbers

1. Series with real terms

Def. A series $\sum_{n=1}^{\infty} a_n$ is called absolutely convergent if the series $\sum_{n=1}^{\infty} |a_n|$ is convergent.

Th 1. Every absolutely convergent series is convergent.

Def. A convergent series $\sum_{n=1}^{\infty} a_n$ which is not absolutely convergent is called a semiconvergent series.

ex. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \ln 2$ is convergent, but $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n}$ is divergent. Hence it is a semiconv. seri

Th 2. (Abel - Dirichlet test) Let (a_n), $n \geq 1$ be a monotonically decreasing sequence with $\lim a_n = 0$ and (b_n), $n \geq 1$ a sequence such that the series $\sum_{n=1}^{\infty} b_n$ has a bounded sequence of partial sums. Then the series $\sum_{n=1}^{\infty} a_n \cdot b_n$ is convergent.

Ex. $\sum_{n=1}^{\infty} \frac{\sin n}{\sqrt{n}}$, take $a_n = \frac{1}{\sqrt{n}} \searrow 0$ and $b_n = \sin n$

$$S_n = b_1 + b_2 + \ldots + b_n = \sin 1 + \sin 2 + \ldots + \sin n$$

$$\sin x + \sin 2x + \sin 3x + \ldots + \sin nx = \frac{\sin \frac{nx}{2} \cdot \sin \frac{(n+1)x}{2}}{\sin \frac{x}{2}}$$

$x \in \mathbb{R} \setminus \{2k\pi \mid k \in \mathbb{Z}\}$

Let $x = 1 \Rightarrow S_n = \sin 1 + \dots + \sin n = \dfrac{\sin\frac{n}{2}\sin\frac{(n+1)}{2}}{\sin\frac{1}{2}}$

$\in \left[-\dfrac{1}{\sin\frac{1}{2}}, \dfrac{1}{\sin\frac{1}{2}}\right] \Rightarrow \sum_{n=1}^{\infty} \dfrac{\sin n}{\sqrt{n}}$ is conv.

Def. A series of the form $a_1 - a_2 + a_3 - a_4 + \dots = \sum_{n=1}^{\infty} (-1)^{n-1} a_n$, $a_n > 0$, $\forall n \geq 1$ is called an alternating series.

Th 3. (Leibniz test) Let (a_n), $n \geq 1$ be a decreasing sequence with $\lim a_n = 0$. Then the alternating series:

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n \text{ is convergent.}$$

2. Properties of series:

Th 1. Let $\sum_{n=1}^{\infty} a_n$ be an absolutely convergent series with sum s. Then for every bijection $\sigma: \mathbb{N}^* \to \mathbb{N}^*$ the series $\sum_{n=1}^{\infty} a_{\sigma(n)}$ is absolutely convergent and its sum is also s. (Permuting terms in an abs. conv. series doesn't change either sum or nature)

Th 2. Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms. Then for every bijection $\sigma: \mathbb{N}^* \to \mathbb{N}^*$ we have.

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_{\sigma(n)}$$

Th 3. Let $\sum_{n=1}^{\infty} a_n$ be a semiconv. series. Then for

every $\alpha \in \overline{\mathbb{R}}$ there exists a permutation $\sigma: \mathbb{N}^* \to \mathbb{N}^*$ s.t.:

$$\sum_{n=1}^{\infty} a_{\sigma(n)} = \alpha \quad \text{(Riemann)}$$

ex. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots = \ln 2$

Rearrange terms $\to 1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12} + \dots$

$$\alpha_{3n} = \sum_{k=1}^{n} \left(\frac{1}{2k-1} - \frac{1}{4k-2} - \frac{1}{4k} \right) = \frac{1}{2} \sum_{k=1}^{n} \left(\frac{1}{2k-1} - \frac{1}{2k} \right)$$

$$= \frac{1}{2}\left(H_{2n} - H_n \right) = \frac{1}{2} \gamma_n - \frac{1}{2} \gamma_n + \frac{1}{2} \ln 2$$

$$\Rightarrow \lim \alpha_{3n} = \frac{\ln 2}{2} \neq \ln 2$$

Def. Let $\sum_{n=0}^{\infty} a_n$, $\sum_{n=0}^{\infty} b_n$ be two series of real numbers. The series $\sum_{n=0}^{\infty} c_n$, where

$$c_n = \sum_{k=0}^{n} a_k b_{n-k} = a_0 b_n + a_1 b_{n-1} + \dots + a_n b_0$$ is called the **Cauchy Product** of the given series.

Obs. In general, the Cauchy product of two conv. series is not necessarily a convergent series.

Th 4. Let $\sum_{n=0}^{\infty} a_n$ be an absolutely conv. series and $\sum_{n=0}^{\infty} b_n$ a convergent series, with $\sum_{n=0}^{\infty} a_n = a$, $\sum_{n=0}^{\infty} b_n = b$. Then the Cauchy product $\sum_{n=0}^{\infty} c_n$ is convergent and $\sum_{n=0}^{\infty} c_n = ab$

Obs. If both series are absolutely convergent, then their Cauchy product is absolutely convergent.

Sequences and series of functions

1. Sequences of functions:

Def. (pointwise convergence) A sequence of functions $(f_n)_{n\geq 1}$, $f_n: A \to \mathbb{R}$ is said to be convergent or pointwise convergent to the function $f: A \to \mathbb{R}$ on A if:

$$\boxed{f(x) = \lim_{n\to\infty} f_n(x), \ \forall x \in A}$$

Denote: $f_n \to f$ or $f_n \xrightarrow{PC} f$

ex. 1) Let $f_n: [0,1] \to \mathbb{R}$, $f(x) = x^n \Rightarrow$

$$\lim_{n\to\infty} f_n(x) = \begin{cases} 0, & x \in [0,1) \\ 1, & x = 1 \end{cases} \Rightarrow f_n \xrightarrow{PC} f, \quad f(x) = \begin{cases} 0, & x \in [0,1) \\ 1, & x = 1 \end{cases}$$

2) Let $f_n: \mathbb{R} \to \mathbb{R}$, $f_n(x) = \dfrac{x + x^2 e^{nx}}{1 + e^{nx}}$, $x \in \mathbb{R}$, $n \in \mathbb{N}^*$

$$\Rightarrow \lim_{n\to\infty} f_n(x) = \begin{cases} x, & x < 0 \\ 0, & x = 0 \\ x^2, & x > 0 \end{cases} \Rightarrow f_n \xrightarrow{PC} f$$

Obs. Pointwise convergence does not generally preserve the properties of the terms of a sequence of functions:

- In example 1, f_n are continous, but the limit is not.

· In example 2, f_n are differentiable, but the limit is not.

Def. (uniform convergence) A sequence of functions $f_n: A \to \mathbb{R}$ is said to be uniformly convergent to $f: A \to \mathbb{R}$ if $\forall \varepsilon > 0$, $\exists n_\varepsilon \in \mathbb{N}$ s.t.:

$$|f_n(x) - f(x)| < \varepsilon, \ \forall n \geq n_\varepsilon \text{ and } x \in A$$

Denote: $f_n \rightrightarrows f$ or $f_n \xrightarrow{UC} f$

Obs. Every uniformly convergent sequence of functions is point wise convergent, but the converse is not generally true.

Th 1. Let (f_n), $n \geq 1$, $f_n: A \to \mathbb{R}$ be a sequence of functions and $f: A \to \mathbb{R}$. If there exists a sequence of positive numbers (a_n), $n \geq 1$ s.t.:

1. $\lim_{n \to \infty} a_n = 0$
2. $|f_n(x) - f(x)| \leq a_n, \ \forall n \geq n_0 \text{ and } x \in A$

$\Rightarrow f_n \xrightarrow{UC} f$ on A

ex. $f_n: \mathbb{R} \to \mathbb{R}$, $f_n(x) = \dfrac{\cos x + 2n \sin x}{n+1}$

$\Rightarrow \lim f_n(x) = 2 \sin x$

$$|f_n(x) - f(x)| = \left|\frac{\cos x - 2 \sin x}{n+1}\right| \leq \frac{3}{n+1}, \ n \geq 1, \ x \in \mathbb{R}$$

$$\lim \frac{3}{n+1} = 0 \Rightarrow f_n \xrightarrow{UC} f$$

Th.2. Let (f_n), $n \geq 1$, $f_n : [a,b] \to \mathbb{R}$ be a sequence of continous functions and $f : [a,b] \to \mathbb{R}$ a continous function s.t.:

1. $f_1(x) \leq f_2(x) \leq f_3(x) \leq \dots$, $\forall x \in [a,b]$
2. $f_n \xrightarrow{pc} f$ on $[a,b]$

$\Rightarrow f_n \xrightarrow{uc} f$ on $[a,b]$ (Dini)

Obs. An essential condition in Dini's theorem is the compactness of the domain of definition of f_n.

Th 3. Let $f_n : A \to \mathbb{R}$, $A \subseteq \mathbb{R}$, be a sequence of continous functions, $f_n \xrightarrow{uc} f$ on A. Then the function is continous on A. (Preservation of continuity)

Th 4. (Passing the limit under the integral) Let (f_n), $n \geq 1$, $f_n : [a,b] \to \mathbb{R}$ be a sequence of continous functions, $f_n \xrightarrow{uc} f$ on $[a,b] \Rightarrow$

$$\lim_{n \to \infty} \int_a^b f_n(x)\,dx = \int_a^b f(x)\,dx$$

Obs. Theorem 4 also holds if (f_n), $n \geq 1$ is a sequence of integrable functions.

Th 5. (Preservation of differentiability) Let (f_n), $n \geq 1$, $f_n : [a,b] \to \mathbb{R}$, be a sequence of differentiable functions s.t.:

1. $f_n \xrightarrow{uc} f$ on $[a,b]$
2. $f_n' \xrightarrow{uc} g$ on $[a,b]$

$\Rightarrow f$ is differentiable on $[a,b]$ and $f' = g$

$$\Rightarrow \left(\lim_{n\to\infty} f_n\right)' = \lim_{n\to\infty} f_n'$$

2. Series of functions:

Def. The series of functions $\sum_{n=1}^{\infty} f_n$ is said to be pointwise (uniformly) convergent if the sequence of its partial sums (s_n), $n \geq 1$ is pointwise (uniformly) convergent

Obs. If $\exists \lim s_n = f \Rightarrow f$ is called the sum of $\sum_{n=1}^{\infty} f_n$.

Denote: $\sum_{n=1}^{\infty} f_n \overset{pc}{=} f$ and $\sum_{n=1}^{\infty} f_n \overset{uc}{=} f$

Th. 6. (Weierstrass) Let $\sum_{n=1}^{\infty} f_n$, $f_n : A \to \mathbb{R}$, be a series of functions and (a_n), $n \geq 1$ a sequence of positive numbers s.t.:

1. $|f_n(x)| \leq a_n$, $\forall n \geq n_0$, $\forall x \in A$
2. $\sum_{n=1}^{\infty} a_n$ is convergent

$\Rightarrow \sum_{n=1}^{\infty} f_n$ is uniformly and absolutely conv. on A

ex. $\sum_{n=1}^{\infty} \frac{\cos nx}{n^2+x^2}$, take $f_n = \frac{\cos nx}{n^2+x^2} \Rightarrow$

$|f_n(x)| \leq \frac{1}{n^2+x^2} \leq \frac{1}{n^2}$ and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is conv. $\Rightarrow \sum_{n=1}^{\infty} f_n$ is unif. and abs. conv.

Th 7. Let (f_n), $n \geq 1$, $f_n : A \to \mathbb{R}$, $A \subseteq \mathbb{R}$ be a sequence of continous functions s.t.:

$$\sum_{n=1}^{\infty} f_n = f \text{ uniformly on } A \Rightarrow f \text{ is a continous function}$$

Th 8 (Termwise differentiation). Let (f_n), $n \geq 1$, $f_n : [a,b] \to \mathbb{R}$, be a sequence of differentiable functions s.t.:

1. $\sum_{n=1}^{\infty} f_n = f$ pointwise on $[a,b]$
2. $\sum_{n=1}^{\infty} f_n' = g$ uniformly on $[a,b]$

$\Rightarrow f$ is differentiable and $f' = g$

$$\Rightarrow \left(\sum_{n=1}^{\infty} f_n\right)' = \sum_{n=1}^{\infty} f_n'$$

Th. 9. (Termwise integration) Let (f_n), $n \geq 1$, $f_n : [a,b] \to \mathbb{R}$, be a sequence of integrable functions s.t. $\sum_{n=1}^{\infty} f_n = f$ uniformly on $[a,b]$. Then f is integrable on $[a,b]$ and:

$$\int_a^b \left(\sum_{n=1}^{\infty} f_n(x)\right) dx = \sum_{n=1}^{\infty} \int_a^b f_n(x)\, dx$$

Th. 10. Let (f_n), $n \geq 1$, $f_n : [a,b] \to [0, +\infty)$, be a neg. of continous nonnegative functions and $f : [a,b] \to \mathbb{R}$ a continous function s.t. $\sum_{n=1}^{\infty} f_n = f$ pointwise on $[a,b]$. Then $\sum_{n=1}^{\infty} f_n$ is uniformly convergent to f on $[a,b]$.